Research Methodology Passport

The Effect of Generative AI-Assisted Programming on Critical Thinking and Academic Performance of Undergraduate IT Students

1. Project information

Title	The Effect of Generative AI-Assisted Programming on Critical Thinking and Academic Performance of Undergraduate IT Students
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GitHub repository	https://github.com/adiirgimbayevv/Methods_Adi_Alan
Based on	Task 1 (research proposal, literature synthesis), refined for Task 2

2. Research type

Quantitative (observational, cross-sectional, correlational with statistical control; quasi-experimental comparison of usage groups). **Justification:** all three research questions ask for a measurable relationship or group difference (critical thinking score, grades, frequency and mode of AI use). Such relationships are tested with regression and group comparison on standardized instruments, which allows hypothesis testing with p-values, confidence intervals and effect sizes. Qualitative interviews are left out of scope to stay feasible within one semester. Because students are not randomly assigned to AI use, the design identifies **associations**, not causal effects (see Limitations).

3. Hypotheses

Primary hypothesis (RQ3, manner of AI use → critical thinking).

H0	After controlling for covariates, there is no difference in critical thinking disposition score between students who mainly use GenAI for explanations/debugging feedback and students who mainly use it for full-solution generation ($\beta_{mode} = 0$).
H1	After controlling for covariates, students who mainly use GenAI for explanations/debugging feedback have a higher critical thinking disposition score than students who mainly use it for full-solution generation ($\beta_{mode} > 0$).

Secondary hypotheses.

ID	H0	H1
RQ1	GenAI use during programming tasks is not associated with critical thinking score ($\beta_{use} = 0$).	GenAI users differ from non-users in critical thinking score ($\beta_{use} \neq 0$).
RQ2	Intensity of GenAI use is not related to academic performance ($\rho = 0$; $\beta_{intensity} = 0$).	Intensity of GenAI use is related to academic performance ($\rho \neq 0$; $\beta_{intensity} \neq 0$).
RQ3b	Manner of use does not affect programming assignment score ($\beta_{mode} = 0$).	Explanation-oriented use gives a higher assignment score than solution generation ($\beta_{mode} > 0$).

4. Variable matrix

Type	Variable	Operationalization
Independent	AI use during coding tasks (RQ1)	Binary: used / did not use GenAI during the task (self-report + task log)
Independent	Intensity of AI use (RQ2)	Sessions per week and share of AI-assisted code steps (0–100%)
Independent	Manner of AI use (RQ3)	Categorical: explanation/debugging mode vs. full-solution generation mode (questionnaire, optionally coded from prompt logs)
Dependent	Critical thinking (disposition)	Total score on the Sosu Critical Thinking Disposition Scale (CTDS, Sosu 2013), a short self-report scale; item-level answers are stored to compute reliability. Configured in *config.yaml*
Dependent	Academic performance	Programming assignment score, practical test score, course final grade (0–100)
Controlled	Student background	Baseline GPA, prior programming experience, year of study, initial subject knowledge

Type	Variable	Operationalization
Controlled	Task conditions	Time spent on task, task difficulty level, individual study time

5. Metrics and baseline

Item	Definition
Primary metric	Adjusted coefficient β_{mode} from OLS regression of CTDS score on usage mode with all controls, reported with 95% CI, one-sided p-value ($\alpha = 0.05$) and standardized effect size (Cohen's d). The hypothesis H1 is supported if $\beta_{mode} > 0$ and $p < 0.05$.
Secondary metrics	Adjusted β for assignment score and final grade (RQ3b, RQ2); Pearson/Spearman ρ for intensity vs. performance; β_{use} for RQ1. Holm correction for multiple tests.
Guardrail metrics	(1) Reliability of the CTDS in our sample, Cronbach's $\alpha \geq 0.70$ (item-level data required); (2) missing data < 10% per variable; (3) multicollinearity VIF < 5; (4) sample size $n \geq$ about 100 (a priori power analysis: $f^2 = 0.15$, $\alpha = 0.05$, power = 0.80); (5) regression assumptions checked (residual normality, homoscedasticity); (6) zero personal data stored: all records anonymized, participation voluntary.
Baseline / control group	(a) **Reference group:** students with no or minimal GenAI use (RQ1) and students using AI for full-solution generation (RQ3); (b) **Null model:** regression with controls only (GPA, prior experience, year, study time); the gain of the full model is reported as ΔR^2 with an F-test, so the AI variables must explain variance beyond background factors.

6. Aim and tasks

Aim. To examine how the frequency and manner of generative AI use in programming courses are linked to critical thinking and academic performance of undergraduate IT students.

Tasks. (1) Review literature on GenAI in programming education, critical thinking and performance. (2) Design measurement of AI-use intensity and mode, critical thinking and academic performance. (3) Collect anonymized survey, test and grade data and prepare a reproducible analysis pipeline. (4) Test H0/H1 for RQ1–RQ3 with controls. (5) Draw conclusions and propose recommendations for GenAI use in programming courses.

7. Limitations and ethics

Causality	Observational design: results show association, not causation. Students with stronger reasoning may choose different AI-use modes (reverse causation); covariates reduce but do not remove confounding.
Measurement	The CTDS measures *disposition* toward critical thinking by self-report, not demonstrated reasoning skill; it is exposed to social-desirability bias. AI-use intensity and mode are also self-reported.
Generalizability	Voluntary participation and one institution limit generalization to other student populations.
Ethics	Participation is voluntary, responses are anonymized, no personal identifiers are stored, and real data are never committed to the repository.

8. Reproducibility note

The repository contains an executable analysis pipeline (*src/main.py*, *configs/config.yaml*, Dockerfile with pinned dependencies) that runs on a synthetic sample (*data/sample/*, 5–10 rows) and writes a result log. Real student data are never committed to the repository.